7^{th} Grade Unit 4 Assessment Guide

General Information	In the 7^{th} grade B.E.S.T. Instructional Guide for Mathematics, a common misconception for the benchmark MA.7.GR.1.1 is "students may incorrectly identify a side length as a height rather than using the perpendicular distance between the bases." The draft test item specifications for 7^{th} grade assessment limit states, "dimensions that are labeled on figures are limited to base(s) and vertical height only." This seems to indicate that within the classroom it is important to guide students to understand the difference between a side and the height but that when students are assessed this common misconception will not be an issue for students. Teachers are cautioned against using this knowledge to eliminate any instruction where the student is learning to distinguish between a side and the height. The unit assessment follows the draft item specs and only labels the base and vertical height for trapezoids and parallelograms. Since there are multiple ways of finding the area of a rhombus using a formula, the unit assessment gives measurements of the diagonals. Another common misconception mentioned in the Instructional Guide is that students may struggle with orientation. As teachers examine the assessment results for all students, teachers are encouraged to note whether the orientation of a figure made a difference in student success. Another thing to consider is whether students struggle with not having a diagram. At the beginning of the assessment is a box with the statements: *Because you are using a calculator, it is possible to accidently punch the wrong button. To ensure you earn partial credit, show the operation that you entered into the calculator and the result. It is helpful if you label the operation with information from the word problem.* This suggestion can also help a student self-scaffold a problem. As you consider the point distribution, consider the sub-steps students may take, including that

	not every student needs to show sub-steps. Your point distribution should not punish students who can think through the problem and find an answer using a calculator such as the Desmos scientific calculator. It is possible that students can enter these problems as one expression into a calculator such as the Desmos scientific calculator to get the correct answer.
Calculator Information	The questions on this assessment are calculator.
Total Recommended Points	The total recommended points in this guide is 100 points.

Question	1	Benchmark(s)	MA.7.GR.1.1
752.64 sq m		**Recommended Scoring** **(10 Total Points)**	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	**2**	**Benchmark(s)**	MA.7.GR.1.3
The relationship between the ○ radius ○ diameter and the ○ area ○ circumference is proportional because the ratio ○ $\frac{A}{d}$ ○ $\frac{C}{d}$ ○ $\frac{A}{r}$ ○ $\frac{C}{r}$ is always the same and can be represented by π.		**Recommended Scoring** **(10 Total Points)**	No partial credit is suggested.
Question	**3**	**Benchmark(s)**	MA.7.GR.1.1
846.72 sq cm		**Recommended Scoring** **(10 Total Points)**	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.

<table>
<tr><td>Question</td><td>4</td><td>Benchmark(s)</td><td>MA.7.GR.1.2</td></tr>
<tr><td colspan="2">5424 sq cm</td><td>Recommended Scoring

(10 Total Points)</td><td>Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.</td></tr>
<tr><td>Question</td><td>5a</td><td>Benchmark(s)</td><td>MA.7.GR.1.3</td></tr>
<tr><td colspan="2" rowspan="2">232.6 or 232.5 or 232.4 in.</td><td>Recommended Scoring

(10 Total Points)</td><td>Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.</td></tr>
<tr><td>Additional Notes for 5a and 5b</td><td>The question does not specify which approximation of pi to use. Students could use the π button on their calculator or 3.14, $\frac{355}{113}$, or $\frac{22}{7}$ for approximations. All possible answers using these values are represented on the key</td></tr>
<tr><td>Question</td><td>5b</td><td>Benchmark(s)</td><td>MA.7.GR.1.4</td></tr>
<tr><td colspan="2">4300.8 or 4302.6 or 4298.7 sq in.</td><td>Recommended Scoring

(10 Total Points)</td><td>Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.</td></tr>
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Question	**6**	**Benchmark(s)**	MA.7.GR.1.1
$720{,}063.3$ sq ft		**Recommended Scoring** **(10 Total Points)**	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	**7**	**Benchmark(s)**	MA.7.GR.1.4
64π sq mm		**Recommended Scoring** **(10 Total Points)**	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	**8**	**Benchmark(s)**	MA.7.GR.1.2
37809.87 sq in		**Recommended Scoring** **(10 Total Points)**	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	**9**	**Benchmark(s)**	MA.7.GR.1.3
19π yd		**Recommended Scoring** **(10 Total Points)**	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.